

Farzaneh Ghanbari

Computer Science PhD Student

Las Cruces, NM, 88003 | +1 (575) 332-1631 | fgh@nmsu.edu | [GitHub](#) | [Website](#)

Research Interests

Knowledge tracing (KT); AI in education; machine learning; deep learning.

Experience

2025–Present | Doctoral Research, *New Mexico State University, Las Cruces, NM, USA*

- Working on the interpretability of deep knowledge tracing models; current focus is QIKT (AAAI 2023), using the pykt-toolkit framework on the ASSIST2009 dataset
- Running and analyzing experiments on QIKT's interpretability scores; work in progress

Fall 2025–Present | Graduate Teaching Assistant, *New Mexico State University, Las Cruces, NM, USA*

- Discrete Mathematics (Fall 2026); Assembly Language Programming (Spring 2026); Assembly Language Programming and Operating Systems I (Fall 2025)
- Graded assignments and exams, held office hours, and assisted students in lab sessions

2025 | Data Collection Assistant, *Department of Mathematics, New Mexico State University, Las Cruces, NM, USA*

Projects

- grid-rl-comparisons** — Implementation and comparison of four reinforcement learning algorithms (Q-Learning, SARSA, DQN, Double DQN) in a 2D grid environment with obstacles and goal states. Python, Jupyter Notebook.
github.com/farzanehghanbari90/grid-rl-comparisons
- Chesscord-using-AI** — A fork of Chesscord extended with LLM-driven chat and AI helpers. JavaScript.
github.com/farzanehghanbari90/Chesscord-using-AI

Skills

- Languages:** Python, C/C++, Java, MATLAB, HTML5, CSS3, JavaScript
- Machine Learning:** PyTorch, pykt-toolkit, scikit-learn, NumPy, pandas
- Databases:** MySQL, SQL Server
- Tools & Environments:** Linux, conda, Git and GitHub, Docker, LaTeX
- Languages (spoken):** Persian (native), English

Education

2025–Present | PhD in Computer Science, *New Mexico State University, Las Cruces, NM, USA*

- Advisor:** Dr. Christabel Wayllace

2013–2016 | Master of Science, Computer Science (Scientific Computing), *Tarbiat Modares University, Tehran, Iran*

- Thesis: a face recognition method based on sparse representation. **Supervisor:** Dr. Mansour Rezghi

2008–2012 | Bachelor of Science, Computer Science, *Shahid Bahonar University of Kerman, Kerman, Iran*

- Thesis: analysis, design and implementation of a hotel booking system using a software engineering approach. **Supervisor:** Dr. Mohamadreza Omid